

## Ideation Phase

### Define the Problem Statements

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Date          | 11 July 2026                                                        |
| Team ID       |                                                                     |
| Project Name  | <b>Electric Vehicle Performance &amp; Market Analysis Dashboard</b> |
| Maximum Marks | 2 Marks                                                             |

#### PS-1

**I am:** Data Analyst

**I'm trying to:** Analyze electric vehicle performance by comparing top speed, driving range, acceleration, pricing, charging speed, and body styles across different brands.

**But:** The available EV data is spread across multiple CSV files with inconsistent formats, missing values, and non-numeric attributes.

**Because:** Raw data cannot directly provide meaningful insights or support decision-making without proper data cleaning and visualization.

**Which makes me feel:** Challenged because identifying trends and comparing EV models manually is time-consuming and error-prone.

#### PS-2

**I am:** Business Intelligence Analyst

**I'm trying to:** Build an interactive Tableau dashboard that enables users to explore EV performance, compare brands, identify top-performing models, and analyze charging station distribution.

**But:** The dataset contains inconsistent values, different units, text-based numerical fields, and missing records that prevent accurate analysis.

**Because:** Without data preprocessing and interactive dashboards, extracting valuable business insights becomes difficult.

**Which makes me feel:** Inefficient because important trends and market opportunities may remain hidden.

## Problem to Solution Flow

**Raw EV CSV Datasets**



**Data Integration (MySQL)**



**Data Cleaning & Transformation**

- Removed missing values
- Converted text values to numeric
- Standardized units
- Handled duplicates
- Feature engineering



**Interactive Tableau Dashboard & Story**

- Brand-wise Top Speed Analysis
- Price Comparison
- Range Analysis
- Acceleration Distribution
- Charging Station Distribution
- Top 10 EV Models



**Business Insights**

- Identify high-performance EV brands
- Compare pricing and driving range
- Analyze body style performance
- Understand regional charging infrastructure



**Better Decision Making**

